Solution Architecture

Automated Employee Onboarding & Offboarding System

Date	15 October 2026	**Team ID**	SkillWallet — ServiceNow Project
Project Name	Automated Employee Onboarding & Offboarding System	**Maximum Marks**	4 Marks

Solution Architecture

Solution architecture bridges the gap between business problems and technology solutions. It defines the structure, characteristics, behavior, and components of the software to stakeholders — including features, development phases, and solution requirements.

Architecture at a glance
A cloud-based ServiceNow solution that uses a Service Catalog request, Flow Designer orchestration, manager approval, departmental tasks, email notifications, lifecycle tracking, and layered access controls. External integrations are planned future scope.

Document contents

1. Solution architecture diagram · 2. Data flow (step-by-step) · 3. Component table · 4. Architecture characteristics and demarcation · 5. Future interfaces · 6. Key design decisions · 7. Benefits

Solution Architecture Diagram

The diagram separates user access, ServiceNow presentation and application logic, data storage, security controls, and planned external systems.

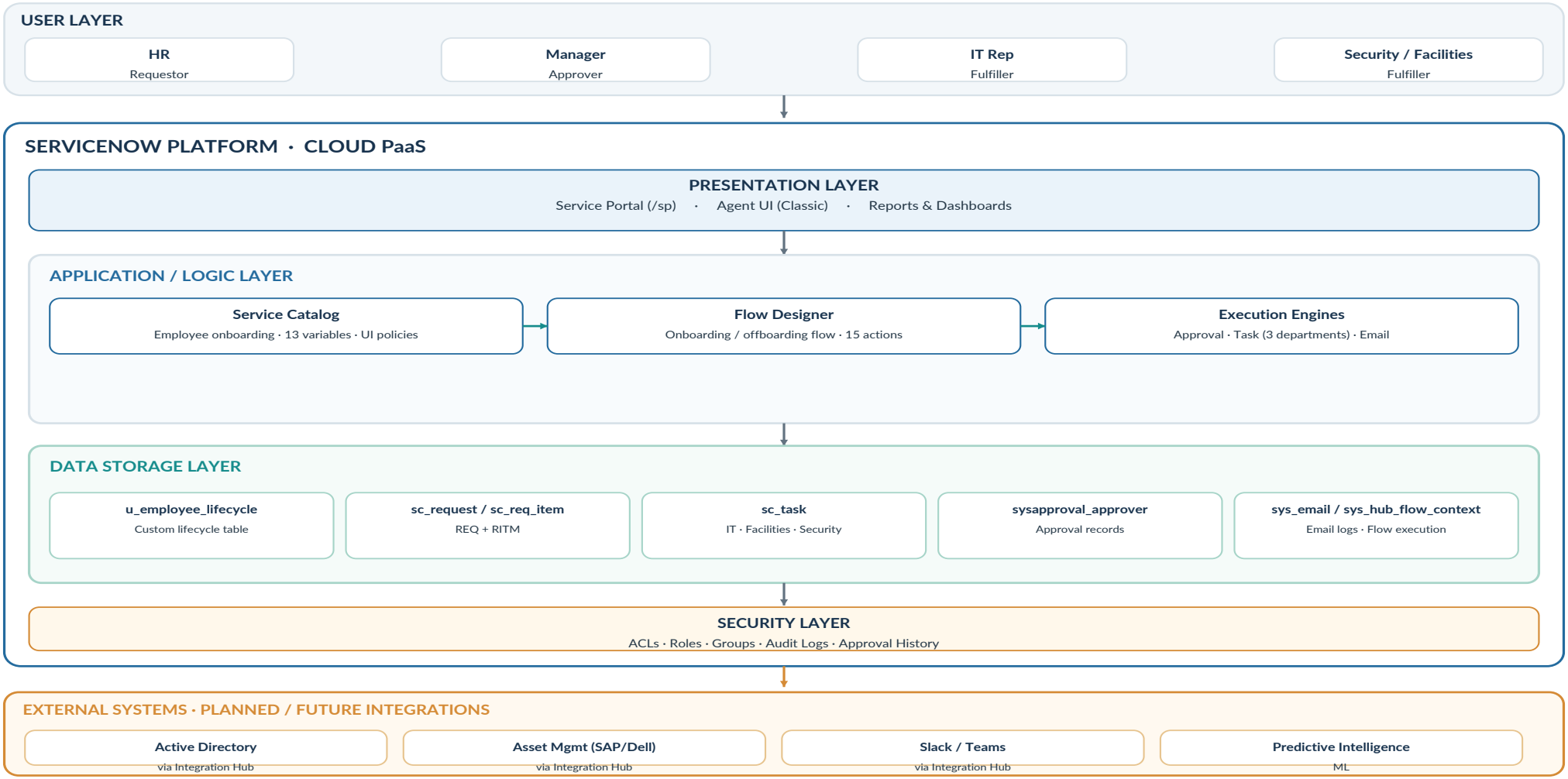
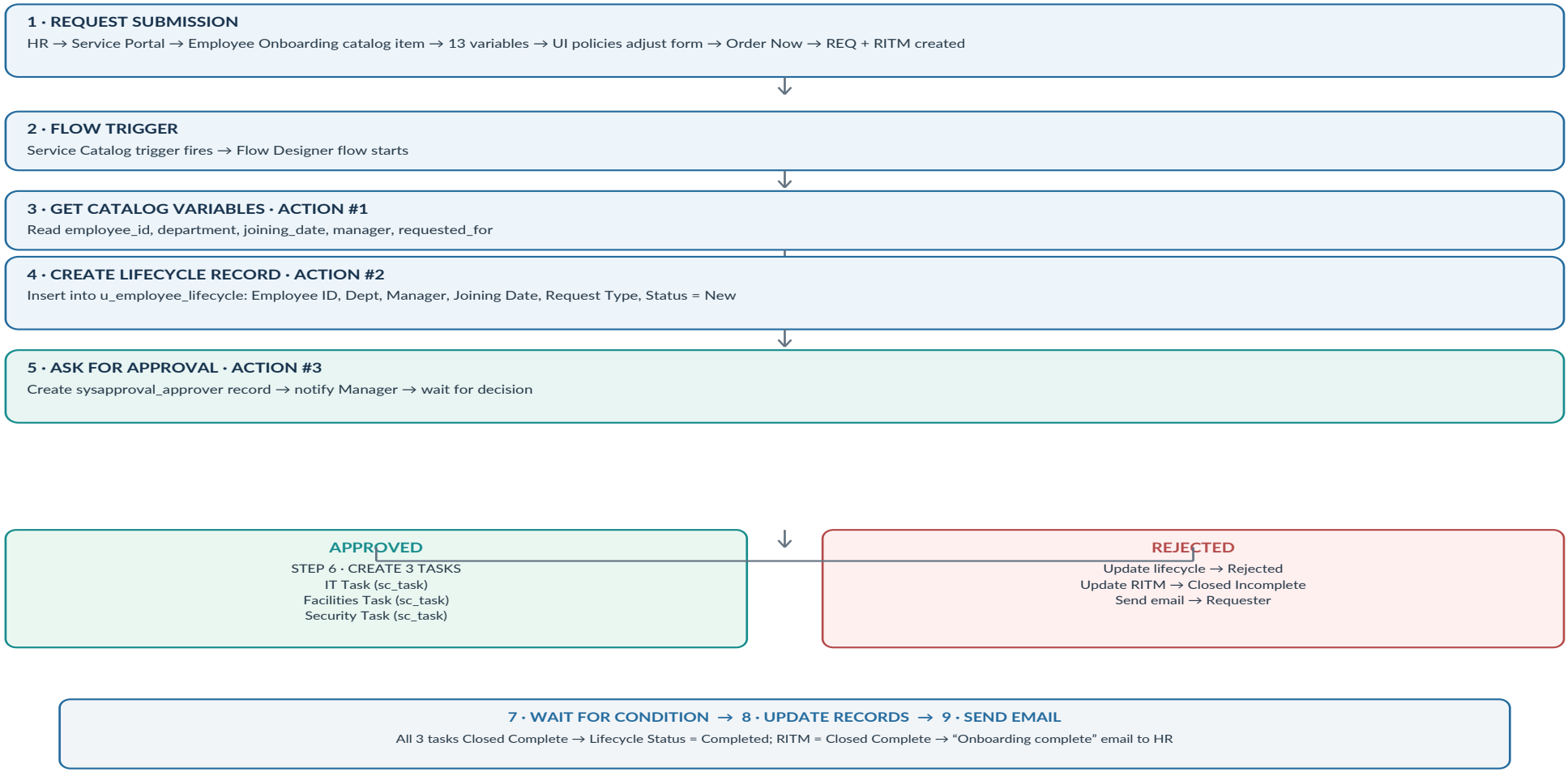


Figure 1: Architecture and data flow of the Automated Employee Onboarding & Offboarding System

Data Flow (Step-by-Step)

The main path runs from request submission through manager approval. Approved requests create three departmental tasks; rejected requests update the lifecycle and request states and notify the requester.



Component Table

S.No	Component	Type	Purpose	Technology
1	Service Portal	Presentation	User-facing request submission	HTML/CSS/JS (ServiceNow)
2	Agent UI	Presentation	Approver/fulfiller interface	ServiceNow Classic UI
3	Service Catalog	Application	Intake form with 13 variables	ServiceNow Catalog
4	Catalog UI Policies	Application	Dynamic field visibility	ServiceNow UI Policies
5	Flow Designer	Application	Automation orchestration	ServiceNow Flow (15 actions)
6	Approval Engine	Application	Manager approval routing	ServiceNow Approvals
7	Task Engine	Application	Auto-task creation	ServiceNow sc_task
8	Notification Engine	Application	Email notifications	ServiceNow Email + SMTP
9	u_employee_lifecycle	Data	Custom lifecycle tracking	MySQL (ServiceNow)
10	sc_request / sc_req_item	Data	REQ + RITM records	MySQL (ServiceNow)
11	sc_task	Data	Departmental tasks	MySQL (ServiceNow)
12	sysapproval_approver	Data	Approval records	MySQL (ServiceNow)
13	sys_email	Data	Email logs	MySQL (ServiceNow)
14	ACLs	Security	Role-based access control	ServiceNow ACL
15	Roles + Groups	Security	Group-to-role mapping	ServiceNow Access
16	Audit Logs	Security	Flow execution + approval history	ServiceNow Audit

Architecture Characteristics

Aspect	Description
Architecture Type	3-tier (Presentation → Application → Data) + Security layer
Deployment Model	Cloud (ServiceNow PaaS)
Automation Level	End-to-end — from request to closure
Custom Code	0 — 100% low-code (Flow Designer)
Data Flow	Unidirectional + conditional branches
Trigger Type	Event-driven (catalog submission)
Error Handling	Flow Error Handler (available)
Audit Trail	Complete (flow execution + approvals + emails)

Architectural Demarcation

Layer	Environment	Components
Presentation	Cloud (ServiceNow)	Portal, Agent UI, Reports
Application	Cloud (ServiceNow)	Catalog, Flow, Approvals, Tasks, Notifications
Data	Cloud (ServiceNow DB)	Custom + OOB tables
Security	Cloud (ServiceNow)	ACLs, Roles, Groups, Audit
External (future)	Cloud (via Integration Hub)	AD, Slack, SAP, Predictive Intelligence

External Interfaces (Future Scope)

Interface	Direction	Purpose	Protocol
Active Directory	Outbound	Auto-create user accounts	LDAP / AD Spoke
Azure AD	Bidirectional	SSO + group sync	OAuth 2.0 / SAML
Slack / Teams	Outbound	Real-time notifications	Webhook / Hub Spoke
SAP / Dell	Outbound	Asset procurement	REST API
Performance Analytics	Internal	Executive dashboards	ServiceNow PA
Predictive Intelligence	Internal	SLA breach prediction	ServiceNow ML

Key Design Decisions

Decision	Reason
Flow Designer over Business Rules	Transparency, admin-maintainable, visual
No Script Includes	Avoid custom code, ease of handoff
Custom table over OOB	Structure data specifically for onboarding/offboarding
Dynamic approver via variable	No hardcoded users; scales with org
Role-based ACLs	Least-privilege data protection
Wait-for-Condition instead of subflows	Simpler, single-flow design
Email as primary notification	Universal, no dependencies

Benefits of This Architecture

Benefit	Impact
Modular	Each layer independent; can swap components
Scalable	Cloud PaaS auto-scales with demand
Secure	Multiple layers of access control
Auditable	Every action traced to source
Extensible	Ready for Integration Hub spokes
Maintainable	100% low-code, admin-editable
Cost-effective	Zero additional software
Reusable	Same pattern for other HR services

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